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DSC 531

Model Selection and Justification

Problem 1:

For the first question, we actually got to our best model 2 separate ways. Both the stepwise selection and forward selection ended up choosing the same best model. That model's metrics (R-Squared, AIC, AICC, MSE, and SBC) were the best of all the models we found. This model had a r-squared of 0.6583 and an AIC value of -3338.29. That marked the highest r-squared value we were able to find. It used 8 of the variables we gave it to sort through. Those were:

1. Hloffer
2. Locale
3. C21enprf
4. Tuition3
5. Fee3
6. Roomcap
7. Roomamt
8. Boardamt

Looking at those selected effects, we can understand why they were selected to be so helpful in predicting graduation rate. Hloffer is for the highest level of degree offered at a certain university and that can point to more prestigious or more academically rigorous universities. Locale represents what type of area the university is and more suburban or city locales could have higher graduation rates than those in more rural areas. C21enprf represents the classification for universities based on the enrollment mix of undergraduate and graduate students. That would be correlated with hloffer because if a university doesn't have any degrees offered above undergraduate, then they wouldn't have a mix of both types of students. The last 5 effects all represent how much money it costs to attend and live at a university. In those cases, a higher amount could indicate a higher graduation rate because those tend to be more selective universities when accepting incoming students. Below is just some of the output from this model to show its metrics:

Effects:	Intercept hloffer locale c21enprf tuition3 fee3 roomcap roomamt boardamt

Analysis of Variance				
Source	DF	Sum of Squares	Mean Square	F Value
Model	24	17.03469	0.70978	72.47
Error	903	8.84359	0.00979	
Corrected Total	927	25.87828		

Root MSE	0.09896
Dependent Mean	0.61476
R-Square	0.6583
Adj R-Sq	0.6492
AIC	-3338.29879
AICC	-3336.74052
SBC	-4147.47299
CV PRESS	9.46996

Problem 2:

For the final best model we found for question 2, we found it through stepwise selection and here were the 8 effects it selected:

1. Outstate
2. Black
3. Men
4. Hisp
5. Twomore
6. Instate
7. White
8. Other

Outstate and instate work the same as they represent what proportion of the students enrolled are from the same state as the university and how many are from a different state. Out of state students theoretically would be more incentivized to graduate because they are paying significantly more than in state students to get the same degree making it a more expensive investment. Then the effects like black, hisp, white, twomore, and other all represent demographical proportions of the university's students. Men represents the proportion of students that are male and males typically have a higher drop out/fail rate than women do in university, so

more men could indicate a lower graduation rate. Below are two tables that show the metrics from our best model. It got our best AIC and SC values of 1741 and 1746 respectively.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	1741.799	817.659
SC	1746.934	863.873
-2 Log L	1739.799	799.659

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	940.1400	8	<.0001
Score	585.4319	8	<.0001
Wald	261.0797	8	<.0001